

IDENTITIES

Q.01. Simplify the following expression

$$\frac{(0.96 \times 0.96 \times 0.96 + 0.04 \times 0.04 \times 0.04)}{(0.96 \times 0.96 - 0.96 \times 0.04 + 0.04 \times 0.04)}$$

- (a) 0 (b) 0.82 (c) 2 (d) 1

Q.02. Simplify:

$$\frac{(379 + 276)^2 + (379 - 276)^2}{379 \times 379 + 276 \times 276}$$

- (a) 2 (b) 655 (c) 103 (d) 1

Q.03. Simplify the given expression.

$$\frac{(381 + 119)^2 + (381 - 119)^2}{(381)^2 + (119)^2}$$

- (a) -1 (b) -2 (c) 1 (d) 2

Q.04. Simplify the given expression

$$\frac{(4.2)^3 - 0.008}{(4.2)^2 + 0.84 + 0.04}$$

- (a) 4 (b) -2 (c) 2 (d) -4

Q.05. Simplify the expression:

$$\frac{143 \times 143 + 143 \times 139 + 139 \times 139}{143 \times 143 + 143 - 139 \times 139 + 139}$$

- (a) $\frac{1}{2}$ (b) 282 (c) $\frac{1}{4}$ (d) 4

Q.06. Simplify the given expression.

$$\frac{(80 \times 80 \times 80) + (70 \times 70 \times 70) + (50 \times 50 \times 50) - 840000}{6400 + 4900 + 2500 - 5600 - 3500 - 4000}$$

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- (a) 100 (b) 200 (c) 400 (d) 300

Q.07. Simplify the given expression

$$\frac{4[(17)^3 - (7)^3]}{(17)^2 + (7)^2 + 119}$$

- (a) 40 (b) 50 (c) 60 (d) 30

Q.08. Simplify:

$$\frac{(612 + 388)^2 + (612 - 388)^2}{(612 \times 612) + (388 \times 388)} = ?$$

- (a) 1 (b) 4 (c) 2 (d) 3

Q.09. Simplify the given expression.

$$\frac{(326 + 222)^2 - (326 - 222)^2}{(326 \times 222)}$$

- (a) 1 (b) 4 (c) 3 (d) 2

Q.10. Simplify the given expression

$$\frac{[120 \times 120 \times 120 - 100 \times 100 \times 100]}{[120 \times 120 + 120 \times 100 + 100 \times 100]}$$

- (a) 20 (b) 25 (c) 10 (d) 15

Q.11. $\frac{5.75 \times 5.75 \times 5.75 + 3.25 \times 3.25 \times 3.25}{57.5 \times 57.5 + 32.5 \times 32.5 - 57.5 \times 32.5}$ is equal to:

- (a) 0.009 (b) 0.0009 (c) 0.9 (d) 0.09

Q.12. Simplify $52 \times 52 - 48 \times 48$.

- (a) 386 (b) 400 (c) 424 (d) 326

Q.13. What is the value of

$$\sqrt{0.52 \times 0.52 + 0.31 \times 0.31 + 0.62 \times 0.52}$$

- (a) 0.63 (b) 0.73 (c) 0.83 (d) 0.93

Q.14. The value of

$$\frac{(a-b)^3 + (b-c)^3 + (c-a)^3}{(a^2-b^2)^3 + (b^2-c^2)^3 + (c^2-a^2)^3}$$
 is equal to

- (a) $ab + bc + ca$ (b) 0
(c) 1 (d) $\frac{1}{(a+b)(b+c)(c+a)}$

Q.15. Simplify:

$$\frac{(r-s)^3 + (s-t)^3 + (t-r)^3}{6(r-s)(s-t)(t-r)}$$

- (a) $\frac{1}{6}$ (b) $\frac{1}{2}$ (c) $\frac{1}{3}$ (d) $\frac{1}{4}$

Q.16. Simplify the expression

$$(36p^2 + 49q^2)(6p + 7q)(6p - 7q)$$

- (a) $1296p^4 + 2401q^4$ (b) $36p^4 - 49q^4$
(c) $1296p^4 - 2401q^4$ (d) $36p^4 + 49q^4$

Q.17. What is the value of $100^2 - 99^2 + 98^2 - 97^2 + 96^2 - 95^2 + 94^2 - 93^2 + \dots + 12^2 - 11^2$?

- (a) 5050 (b) 4985 (c) 4995 (d) 4950

Q.18. Simplify the given expression.

$$(5x + 8y)(25x^2 + 64y^2 - 40xy)$$

- (a) $125x^3 - 512y^3$ (b) $25x^3 + 64y^3$
(c) $125x^3 + 512y^3$ (d) $25x^3 - 64y^3$

Q.19. What is the value of

$$\frac{(91)^3 + (09)^3}{[(91)^2 - 0.819 + (09)^2]}$$

- (a) 1 (b) 5 (c) 4 (d) 6

Q.20. The value of

$$\frac{(0.545)(0.081)(0.51)(5.2)}{(0.324)^3 + (0.221)^3 - (0.545)^3}$$
 is:

- (a) -1 (b) 1 (c) 3 (d) -3

Q.21. The value of

$$\frac{(4.6)^4 + (5.4)^4 + (24.84)^2}{(4.6)^2 + (5.4)^2 + 24.84}$$
 is:

- (a) 24.42 (b) 24.24 (c) 25.42 (d) 25.48

Q.22. Simplify the following expression.

$$(2x - 3y)^3 - 18xy(2x - 3y)$$

- (a) $8x^3 - 27y^3 - 36x^2y - 54xy^2$
(b) $8x^3 - 27y^3 - 72x^2y + 108xy^2$
(c) $8x^3 - 27y^3$
(d) $8x^3 + 108xy^2 - 72x^2y$

Q.23. Simplify the following expressions:

$$\frac{(a^2 - 4b^2)^3 + 64(b^2 - 4c^2)^3 + (16c^2 - a^2)^3}{(a - 2b)^3 + (2b - 4c)^3 + (4c - a)^3}$$

- (a) $-(a - 2b)(b + 2c)(4c + a)$
(b) $2(a + 2b)(b + 2c)(4c + a)$
(c) $(a + 2b)(b + 2c)(4c + a)$
(d) $4(a + 2b)(b + 2c)(4c + a)$

Q.24. Simplify $(5 \times 5 \times 5 \times 5 \times 5)^5 \times (5 \times 5 \times 5)^5 \div 5 = (125)^?$

- (a) 15 (b) 21 (c) 13 (d) 14

Q.25. If $847 \times 385 \times 675 \times 3025 = 3^a \times 5^b \times 7^c \times 11^d$, then the value of $ab - cd$ is:

- (a) 5 (b) 1 (c) 4 (d) 7

Q.26. Simplify: $(a^{-1} + b^{-1}) \div (a^{-3} + b^{-3})$

- (a) $\frac{ab}{a^2 - ab + b^2}$ (b) $\frac{a^3 b^3}{a^2 - ab + b^2}$
(c) $\frac{a^2 b^2}{a^2 + ab + b^2}$ (d) $\frac{a^2 b^2}{a^2 - ab + b^2}$

Q.27. Simplify the following expression.

$$\frac{a^3 + b^3 + c^3 - 3abc}{a^2 + b^2 + c^2 - ab - bc - ca}$$

Where; $a = 2.01, b = 39.75, c = 28.24$

- (a) 140 (b) 50 (c) 70 (d) 100

Q.28. If a, b, c are positive real numbers, what is the value of

$$\sqrt{a^{-1}b} \times \sqrt{b^{-1}c} \times \sqrt{c^{-1}a}$$

- (a) 2 (b) 3 (c) 1 (d) 4

Q.29. Find the value of

$$\left(\frac{9^3 + 9^4 + 9^5 + 9^6 + 9^7}{9^1 + 9^2 + 9^3 + 9^4 + 9^5} \right)^{\frac{1}{2}} = ?$$

- (a) 9 (b) 3 (c) 729 (d) 81

Q.30. Simplify the following expression.

$$(2.19)^2 + 2(2.19)(3.81) + (3.81)^2$$

- (a) 216 (b) 12 (c) 36 (d) 6

Q.31. If $A = \frac{58^2 - 25^2}{46^2 - 37^2}$, $B = \frac{26^2 - 15^2}{56^2 - 15^2}$,

then the value of $\frac{1}{B} - \frac{20}{A}$ is:

- (a) 2 (b) 1 (c) -1 (d) 0

Q.32. Simplify:

$$\frac{(3.321)^2 + (2.681)^2 + (1.245)^2 - 3 \times 3.321 \times 2.681 \times 1.245}{(3.321)^2 + (2.681)^2 + (1.245)^2 - (3.321 \times 2.681) - (2.681 \times 1.245) - (1.245 \times 3.321)}$$

- (a) 6.125 (b) 8.645 (c) 7.247 (d) 10.245

BASED ON SQUARE ROOTS/ CONJUGATE/SURDS

Q.33. Simplify the given expression.

$$\frac{0.09 \times 0.09 + 0.04 \times 0.16 + 2 \times 0.09 \times 0.04 + 2 \times 0.04 \times 0.16 + 2 \times 0.09 \times 0.16}{0.3 \times 0.3 + 0.2 \times 0.2 + 0.4 \times 0.4}$$

- (a) 0.38 (b) 0.24 (c) 0.32 (d) 0.29

Q.34. Solve

$$\frac{(6.9)^3 + (2.7)^3 + (5.8)^3 - 3 \times 6.9 \times 2.7 \times 5.8}{(6.9)^2 + (2.7)^2 + (5.8)^2 - (6.9)(2.7) - (2.7)(5.8) - (5.8)(6.9)}$$

- (a) 15.1 (b) 15.3 (c) 15.2 (d) 15.4

Q.35. Find the value of: $13^3 + (-8)^3 + (-5)^3$

- (a) 1052 (b) 1152 (c) 1562 (d) 1560

Q.36. Find the value of $\frac{\sqrt{1.24} \times \sqrt{2.79}}{\sqrt{2.64} \times \sqrt{5.94}}$

- (a) $\frac{31}{66}$ (b) $\frac{31}{44}$ (c) $\frac{33}{64}$ (d) $\frac{33}{31}$

Q.37. Find the value of

$$\sqrt{625} + \sqrt{\frac{625}{100}} + \sqrt{\frac{625}{10000}} + \sqrt{\frac{625}{1000000}}$$

- (a) 27.075 (b) 27.775
(c) 27.75 (d) 27.025

Q.38. The expression

$$\frac{15(\sqrt{10} + \sqrt{5})}{\sqrt{10} + \sqrt{20} + \sqrt{40} - \sqrt{5} - \sqrt{80}}$$
 is equal to:

- (a) $5 + 2\sqrt{2}$ (b) $5 - 2\sqrt{5}$
(c) $5(3 + 2\sqrt{2})$ (d) $10(3 + 2\sqrt{5})$

Q.39. The value of

$$\sqrt{28 + 10\sqrt{3}} - \sqrt{7 - 4\sqrt{3}}$$
 is closest to:

- (a) 7.2 (b) 6.1 (c) 6.5 (d) 5.8

Q.40. If $\sqrt{11 - 3\sqrt{8}} = a + b\sqrt{2}$, then what is the value of $(2a + 3b)$?

- (a) 5 (b) 7 (c) 9 (d) 3

Q.41. If $x = \sqrt{-\sqrt{3} + \sqrt{3} + 8\sqrt{7} + 4\sqrt{3}}$ where $x > 0$, then the value of x is equal to:

- (a) 2 (b) 3 (c) 4 (d) 1

Q.42. Find the value of x .

$$\sqrt[3]{15625} - \sqrt{x} = 4$$

- (a) 625 (b) 343 (c) 441 (d) 81

Q.43. Let $x = \left(\frac{\sqrt{1875}}{\sqrt{3888}} \div \frac{\sqrt{1200}}{\sqrt{768}}\right) \times$

$$\frac{\sqrt{175}}{\sqrt{1792}}$$
 Then $\sqrt{x}$ is equal to:

- (a) $\frac{7}{12}$ (b) $\frac{4}{9}$ (c) $\frac{5}{12}$ (d) $\frac{5}{9}$

Q.44. If $(\sqrt{2} + \sqrt{5} - \sqrt{3}) \times k = -12$, then what will be the value of k ?

- (a) $(\sqrt{2} + \sqrt{5} + \sqrt{3})$
(b) $(\sqrt{2} + \sqrt{5} + \sqrt{3})(2 - \sqrt{10})$
(c) $(\sqrt{2} + \sqrt{5} - \sqrt{3})(2 + \sqrt{5})$
(d) $(\sqrt{2} + \sqrt{5} + \sqrt{3})(2 - \sqrt{5})$

Q.45. The expression

$$\sqrt{10 + 2(\sqrt{6} - \sqrt{15} - \sqrt{10})}$$
 is equal to

- (a) $\sqrt{3} + \sqrt{2} - \sqrt{5}$ (b) $\sqrt{3} - \sqrt{2} - \sqrt{5}$
(c) $\sqrt{3} - \sqrt{2} + \sqrt{5}$ (d) $\sqrt{2} - \sqrt{3} - \sqrt{5}$

Q.46. If $5\sqrt{3} + \sqrt{75} = 17.32$, then the value of $14\sqrt{3} + \sqrt{108}$ is:

- (a) 32.46 (b) 35.64 (c) 34.64 (d) 33.86

Q.47. Find the value of $\sqrt{2 + \sqrt{3}} + \sqrt{2 - \sqrt{3}}$.

- (a) $\sqrt{6}$ (b) $2\sqrt{3}$ (c) $2\sqrt{2}$ (d) 6

Q.48. Find the value of the given expression.

$$\sqrt{20} - \sqrt{20} - \sqrt{20} - \sqrt{20} - \dots \dots \dots \infty$$

- (a) 4 (b) 6 (c) 5 (d) 2

Q.49. Simplify the following.

$$\frac{\sqrt{10 + \sqrt{25 + \sqrt{108 + \sqrt{154 + \sqrt{225}}}}}}{\sqrt{16 + 19.25 \times 4^2}}$$

- (a) $\frac{7}{18}$ (b) $\frac{2}{9}$ (c) $\frac{5}{18}$ (d) $\frac{1}{9}$

Q.50. Simplify $\sqrt{36x^2 - 108x + 81}$.

- (a) $6x - 9$ (b) $2x - 9$ (c) $5x - 9$ (d) $3x - 3$

Q.51. What is the value of

$$\frac{\sqrt{29.16}}{\sqrt{1.1664}} + \frac{\sqrt{0.2916}}{\sqrt{116.64}} + \frac{\sqrt{0.0036}}{\sqrt{0.36}}?$$

- (a) $\frac{26}{5}$ (b) $\frac{103}{20}$ (c) $\frac{27}{5}$ (d) $\frac{101}{20}$

Q.52. If $A = \frac{\sqrt{0.0004} \times \sqrt[3]{0.000008}}{\sqrt[4]{16000} \times \sqrt[3]{125000} \times \sqrt[4]{810}}$ and $B = \frac{\sqrt[3]{0.729} \times \sqrt[3]{0.0016}}{\sqrt{0.16}}$, then what is $A \times B$?

- (a) 7×10^{-7} (b) $(\frac{7}{4}) \times 10^{-8}$
(c) $\frac{3}{5} \times 10^{-7}$ (d) $(\frac{7}{3}) \times 10^{-7}$

Q.53. If $\sqrt{5} = 2.24$, then find the value of

$$\sqrt{\frac{\sqrt{5} - 1}{\sqrt{5} + 1}}$$

- (a) 0.62 (b) 0.59 (c) 0.68 (d) 0.55

Q.54. On simplifying the following expression, we get _____.

$$\frac{24\sqrt{45} - 16\sqrt{20} + \sqrt{245} - 47\sqrt{5}}{3\frac{1}{5} + 1\frac{2}{5} - 2\frac{1}{6} - 2\frac{1}{5} - 5\frac{2}{5} + 3\frac{1}{6}}$$

- (a) 0 (b) 3 (c) 1 (d) 2

Q.55. Evaluate $\frac{\sqrt{3} + \sqrt{2}}{\sqrt{3} - \sqrt{2}}$, given that

$$\sqrt{6} = 2.45$$

- (a) 7.7 (b) 9.9 (c) 8.8 (d) 6.6

Q.56. Simplify the expression $\frac{\sqrt{x} - \sqrt{y}}{\sqrt{x} + \sqrt{y}}$, where $x = 2$ and $y = 3$.

- (a) $2\sqrt{6} - 6$ (b) $\sqrt{6} - 5$
(c) $5 - 2\sqrt{6}$ (d) $2\sqrt{6} - 5$

Q.57. The value of

$$\frac{7 + 3\sqrt{5}}{3 + \sqrt{5}} - \frac{7 - 3\sqrt{5}}{3 - \sqrt{5}}$$
 lies between:

- (a) 2 and 2.5 (b) 3 and 3.5
(c) 1.5 and 2 (d) 2.5 and 3

Q.58. The value of

$$\frac{2\sqrt{10}}{\sqrt{5} + \sqrt{2} - \sqrt{7}} - \sqrt{\frac{\sqrt{5} - 2}{\sqrt{5} + 2}} - \frac{3}{\sqrt{7} - 2}$$
 is:

- (a) $2 + \sqrt{2}$ (b) $2\sqrt{5}$ (c) $\sqrt{2}$ (d) $\sqrt{7}$

Q.59. The value of

$$5 - \frac{8 + 2\sqrt{15}}{4} - \frac{1}{8 + 2\sqrt{15}}$$
 is equal to:

- (a) $\frac{2}{3}$ (b) 1 (c) $\frac{1}{2}$ (d) $\frac{1}{4}$

Q.60. The value of

$$\frac{1}{4 - \sqrt{15}} - \frac{1}{\sqrt{15} - \sqrt{14}} + \frac{1}{\sqrt{14} - \sqrt{13}} - \frac{1}{\sqrt{13} - \sqrt{12}} + \frac{1}{\sqrt{12} - \sqrt{11}} - \frac{1}{\sqrt{11} - \sqrt{10}} + \frac{1}{\sqrt{10} - 3} - \frac{1}{3 - \sqrt{8}}$$
 is:

- (a) $4 - 2\sqrt{2}$ (b) $2 + 2\sqrt{2}$
(c) $4 + 2\sqrt{2}$ (d) $2 - 2\sqrt{2}$

Q.61. If $\frac{\sqrt{38 - 5\sqrt{3}}}{\sqrt{26 + 7\sqrt{3}}} = \frac{a + b\sqrt{3}}{23}$, $b > 0$, then the value of $(b - a)$ is:

- (a) 7 (b) 18 (c) 29 (d) 11

Q.62. If $\frac{1}{4 - \sqrt{8}} + \frac{3 + 2\sqrt{2}}{3 - 2\sqrt{2}} - \frac{3 - 2\sqrt{2}}{3 + 2\sqrt{2}} = a + b\sqrt{2}$, then what is the value of $(3a + 4b)$?

- (a) $99\frac{1}{2}$ (b) 97 (c) $98\frac{1}{2}$ (d) 98

Q.63. If $\frac{4}{1 + \sqrt{2} + \sqrt{3}} = a + b\sqrt{2} + c\sqrt{3} - d\sqrt{6}$, where a, b, c, d are natural numbers, then the value of $a + b + c + d$ is:

- (a) 0 (b) 2 (c) 4 (d) 1

Q.64.

$$\text{If } \left[\left(\frac{2}{3}\right)^3\right]^{2x+3} \cdot \frac{-3}{4} = \left[\left(\frac{2}{3}\right)^{\frac{2}{3}}\right]^{3x+7} \cdot \frac{-6}{5}$$

then the value of $\sqrt{2 - 42x}$ is:

- (a) 5 (b) 6 (c) 3 (d) 4